

Task 01

Network Scanning & Reconnaissance Report

Introduction

This report documents a network scan performed using Nmap to identify open ports, services and operating system information. The scan was performed on a personal system in local network environment using Nmap commands. This task helped in understanding the basics of network reconnaissance and security analysis.

Tools Used

- Nmap (Network Mapper)
- Zenmap
- Windows Command Prompt
- Operating System: Windows 11

Findings

Active Host:

10.185.57.xxx

Open Ports:

- 135 (MSRPC)
- 139 (NetBIOS)
- 445 (SMB / Microsoft-DS)

Operating System Detected:

- Microsoft Windows 11

Risk Analysis

Port 135:

Used for Microsoft RPC communication. If exposed publicly, attackers may misuse vulnerabilities.

Port 139:

Used for NetBIOS file sharing service. May expose system information on network.

Port 445:

Used for SMB file sharing. Commonly targeted in malware and ransomware attacks.

Recommendations

- Keep Windows updated
- Enable firewall protection
- Disable unnecessary sharing services
- Use antivirus protection
- Avoid exposing SMB services publicly

Scan Results Explanation

The scan detected:

- One active host
- Multiple open ports
- Windows related services
- Operating system details

Learning Outcome

Through this task, the following concepts were understood:

- Basics of network reconnaissance
- Use of Nmap commands
- Identification of open ports and services

- Understanding basic security risks
- Importance of network security

Conclusion

The network scan successfully identified open ports, running services and operating system information of the target system. This task helped in understanding how reconnaissance scanning is performed using Nmap and how open services may create security risks if not properly secured.

Output

```
C:\Users\>nmap -sV -O 10.185.57. -oN final_report.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 09:10 +0530
Nmap scan report for 10.185.57.
Host is up (0.00077s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
Device type: general purpose
Running: Microsoft Windows 11
OS CPE: cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 11 24H2 - 25H2
Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.36 seconds
```